Exercise 1: Worked solutions

Electrical Engineering I · Module 92315 · Exercise 1

From Fundamentals to AI Apocalypse · personal study guide

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Work each part on paper before opening its solution. Each answer builds from the geometry or field definition, then checks the result.

Task 1: Cylinder in cylindrical coordinates

A right circular cylinder has radius a, height h, and its axis lies along z. The origin is at the centre of the base; z points upward.

1a: Describe the three boundary surfaces

Try first: write the coordinate ranges for the base, curved side, and top.

Answers: see "Task 1: Cylinder in cylindrical coordinates" under Answers at the end.

1b: Give the outward unit normals

Try first: identify which coordinate direction points out of each smooth face.

Answers: see "Task 1: Cylinder in cylindrical coordinates" under Answers at the end.

Task 2: Hemisphere of radius a

Include both the curved half-sphere and its flat base when finding total surface area.

2a: Total surface area

Try first: calculate the curved part and disk separately, then add them.

Answers: see "Task 2: Hemisphere of radius a" under Answers at the end.

2b: Volume

Try first: set the spherical radial limit to a and the polar angle to the upper half.

Answers: see "Task 2: Hemisphere of radius a" under Answers at the end.

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Task 3: Field integral around a circle

Integrate each field around a circle of radius $\rho = b$ centred at the origin. Assume φ increases from 0 to 2π, counterclockwise when viewed from $+z$. Reversing this path reverses any nonzero result.

3a: Uniform x-directed field

Given $H = H_0 e_x$. Try first: parameterize the circle and evaluate the dot product.

Answers: see "Task 3: Field integral around a circle" under Answers at the end.

3b: Radial field

Given $H = H_0 e_\rho$. Try first: compare the field direction with the path tangent.

Answers: see "Task 3: Field integral around a circle" under Answers at the end.

3c: Azimuthal field growing with radius

Given $H = H_0(\rho/a) e_\varphi$. Try first: evaluate the field on $\rho = b$ and use the circle's tangent element.

Answers: see "Task 3: Field integral around a circle" under Answers at the end.

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Answers

Task 1: Cylinder in cylindrical coordinates

1a · Surface coordinates

1. In cylindrical coordinates a point is $r = \rho\, e_\rho + z\, e_z$. A disk varies in ρ and φ while holding z fixed; the side holds ρ fixed.
2. For the base, set $z = 0$: $0 \le \rho \le a$, $0 \le \varphi < 2\pi$. For the top, set $z = h$ with the same disk ranges.
3. For the lateral surface, set $\rho = a$, with $0 \le \varphi < 2\pi$ and $0 \le z \le h$.

   Result: base: $(\rho, \varphi, 0)$; side: (a, φ, z); top: (ρ, φ, h), over the ranges above.

   Check: the curved side has no radial thickness; each disk spans the full radius.

Task 1: Cylinder in cylindrical coordinates

1b · Outward normals

1. The bottom face lies at $z = 0$. The cylinder occupies larger z, so outward points downward: $-e_z$.
2. The side is at fixed $\rho = a$; increasing ρ leaves the cylinder, so its normal is $+e_\rho$.
3. The top is at $z = h$, so outward is $+e_z$. Thus the normals are base $-e_z$, side $+e_\rho$, top $+e_z$. At the sharp circular rims, the adjacent face normals differ, so there is no single smooth-face normal.

   Check: each normal points away from the cylinder's interior.

Task 2: Hemisphere of radius a

2a · Curved surface plus base

1. Use spherical coordinates with polar angle θ measured down from $+z$. On the curved hemisphere, $r = a$, $0 \le \theta \le \pi/2$, $0 \le \varphi < 2\pi$, and $dA = a^2 \sin\theta\, d\theta\, d\varphi$.
2. Integrate the curved part: $A_{curve} = \int_0^{2\pi} \int_0^{(\pi/2)} a^2 \sin\theta\, d\theta\, d\varphi = 2\pi a^2$.
3. The base disk has area $A_{base} = \pi a^2$. Add the two pieces: $\mathbf{A_{total} = 2\pi a^2 + \pi a^2 = 3\pi a^2.}$

   Check: the curved half is half the sphere's $4\pi a^2$ area; remember to include the base.

Task 2: Hemisphere of radius a

2b · Spherical volume integral

1. For the hemisphere, $0 \le r \le a$, $0 \le \theta \le \pi/2$, and $0 \le \varphi < 2\pi$. The volume element is $dV = r^2 \sin\theta \, dr \, d\theta \, d\varphi$.
2. Separate the factors: $V = (\int_0^a r^2 \, dr)(\int_0^{\pi/2} \sin\theta \, d\theta)(\int_0^{2\pi} d\varphi) = (a^3/3)(1)(2\pi)$.
3. **Result: $V = 2\pi a^3/3$.**

   Check: it is half the full sphere's $4\pi a^3/3$ volume.

Task 3: Field integral around a circle

3a: Uniform field

1. On the circle, $ds = b \, e_\varphi \, d\varphi$, where $e_\varphi = -\sin\varphi \, e_x + \cos\varphi \, e_y$.
2. Then $H \cdot ds = -b \, H_0 \sin\varphi \, d\varphi$. Integrate around the full loop: $\oint H \cdot ds = -b \, H_0 \int_0^{2\pi} \sin\varphi \, d\varphi$.
3. **Result: $\oint \boldsymbol{H} \cdot \mathbf{d}\boldsymbol{s} = 0$.** The positive and negative tangential contributions cancel over a full turn.

   Check: the field is not perpendicular everywhere; zero comes from cancellation, not pointwise orthogonality.

Task 3: Field integral around a circle

3b: Radial field

1. The circle's line element is tangential: $ds = b \, e_\varphi \, d\varphi$. The field points radially along e_ρ.
2. Radial and azimuthal unit vectors are perpendicular, so $e_\rho \cdot e_\varphi = 0$ at every point. Hence $H \cdot ds = 0$ everywhere on the path.
3. **Result: $\oint \boldsymbol{H} \cdot \mathbf{d}\boldsymbol{s} = 0$.**

   Check: unlike 3a, this integrand vanishes point by point, so no cancellation argument is needed.

Task 3: Field integral around a circle

3c · Tangential field

1. On $\rho = b$, the field is $H = (b/a)H_0 e_\varphi$; the path element is $ds = b \, e_\varphi \, d\varphi$.
2. The directions agree, so $H \cdot ds = (H_0 b^2/a) d\varphi$. Integrating through 2π gives $2\pi H_0 b^2/a$.
3. **Result: $\oint \boldsymbol{H} \cdot \mathbf{d}\boldsymbol{s} = 2\pi \boldsymbol{H_0 b^2/a}$.** The assumed counterclockwise direction makes it positive.

   Check: H has units A/m, so H times path length has units A. Clockwise traversal changes the sign.

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